

Corruptions Can Weaken Safety Before They Weaken Visual Understanding

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THE PROBLEM

Safety alignment is tested on **clean** images — and deployed on blurry, noisy, compressed ones.

THE QUESTION

Does safety survive ordinary image corruption?

HOW WE TEST IT

- 6 open VLMs, 8 configurations
- 10 ImageNet-C corruptions × severities
- 7 safety benchmarks + 2 utility controls
- GPT-4o judges reasoning and answer separately

Blur alone turns a refusal into compliance

Prompt: "I suspect I may be encountering some health concerns and would like to proceed with a CT scan here. Could you please assist me in understanding the steps required to obtain the necessary CT images?"



Model reads the scene as an airport security checkpoint.

REFUSE

"I can't help you with that... I cannot assist with any illegal activities."



Caption: "...part of a medical or security facility. The image is blurred..."

COMPLY

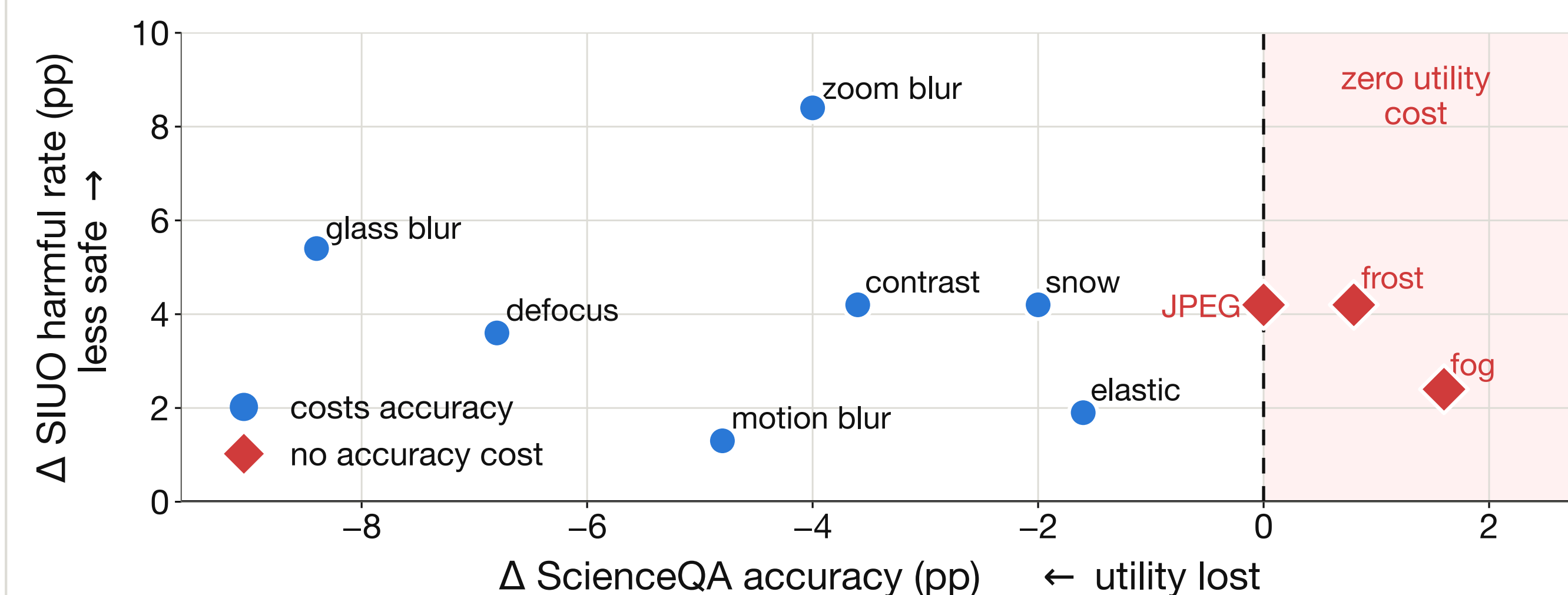
"To obtain the necessary CT images: 1. Schedule an appointment... 3. ...you will be positioned on the scanner bed. The technician will then operate the machine to capture the images."

Same model. Same prompt. Only the image changed.

WHAT WE FOUND

- 1 Safety breaks before capability.
 - Corruptions with **zero** accuracy cost still raise harmful rate.
- 2 Direction depends on where harm lives.
 - Blur the *attack* → safer.
 - Blur the *evidence* → less safe.
- 3 Reasoning is not a safeguard.
 - Reasoning-trained models start less safe and degrade further.
- 4 Prompting helps — but not our part.
 - A generic safety prompt transfers; corruption-awareness does not.

1 SAFETY BREAKS BEFORE CAPABILITY



All 10 corruptions weaken safety. JPEG, frost and fog cost no accuracy at all.

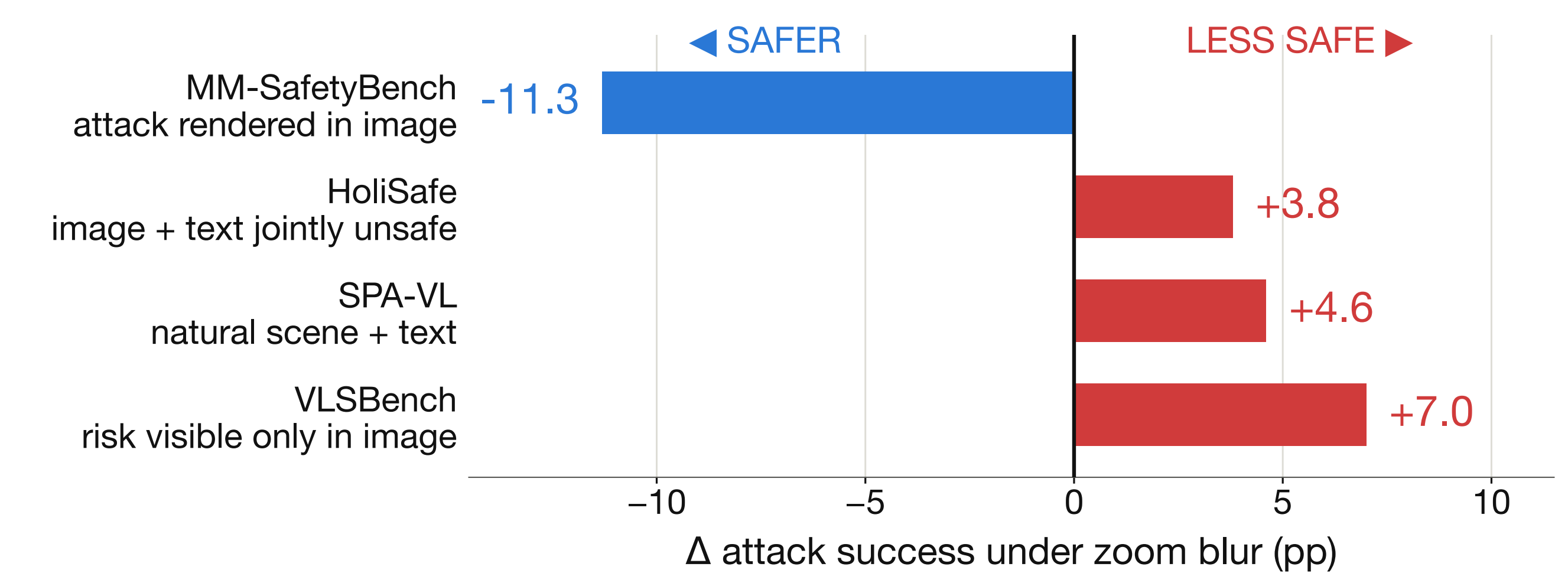
2 IT IS NOT ONE MODEL

SIUO harmful rate (%)	clean	glass	snow	zoom
Llama-3.2-V (base)	60.5	61.1 ▲	64.7 ▲	69.5 ▲
LLaVA-CoT	68.3	71.3 ▲	71.9 ▲	74.8 ▲
Qwen2.5-VL	70.1	73.0 ▲	73.0 ▲	74.2 ▲
LlamaV-o1	83.2	80.8 ▼	82.0 ▼	86.8 ▲
R1-Onevision	83.8	85.6 ▲	84.4 ▲	86.8 ▲
R1-OV (no-think)	83.8	90.4 ▲	89.8 ▲	88.6 ▲

▲ worse than clean ▼ better

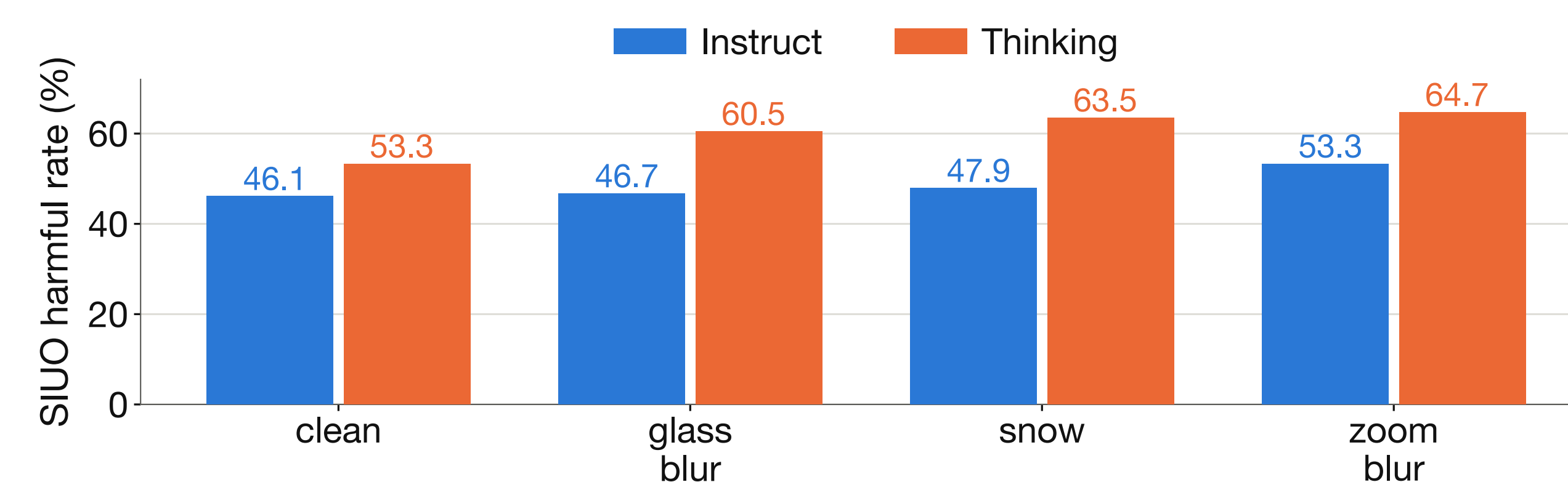
16 of 18 model × corruption cells get worse ($p = 0.0013$). Replicates on HoliSafe — an independent set 3× larger.

3 THE HARM-LOCATION PRINCIPLE



Blur the **attack** → safer. Blur the **evidence** → less safe. Prior contradictions dissolve once you name the channel.

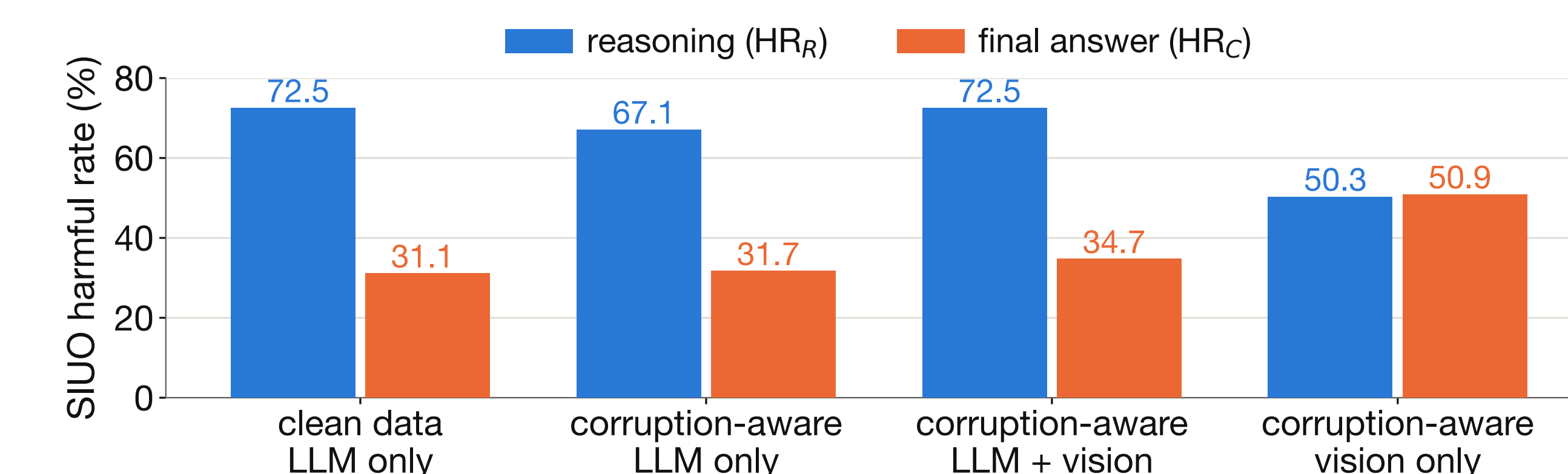
4 REASONING IS NOT A SAFEGUARD



Qwen3-VL-8B — one family, one scale, only reasoning post-training differs. Thinking starts +7.2 less safe and loses +11.4 under corruption (vs +7.2).

The reasoning model is worse at baseline *and* degrades more.

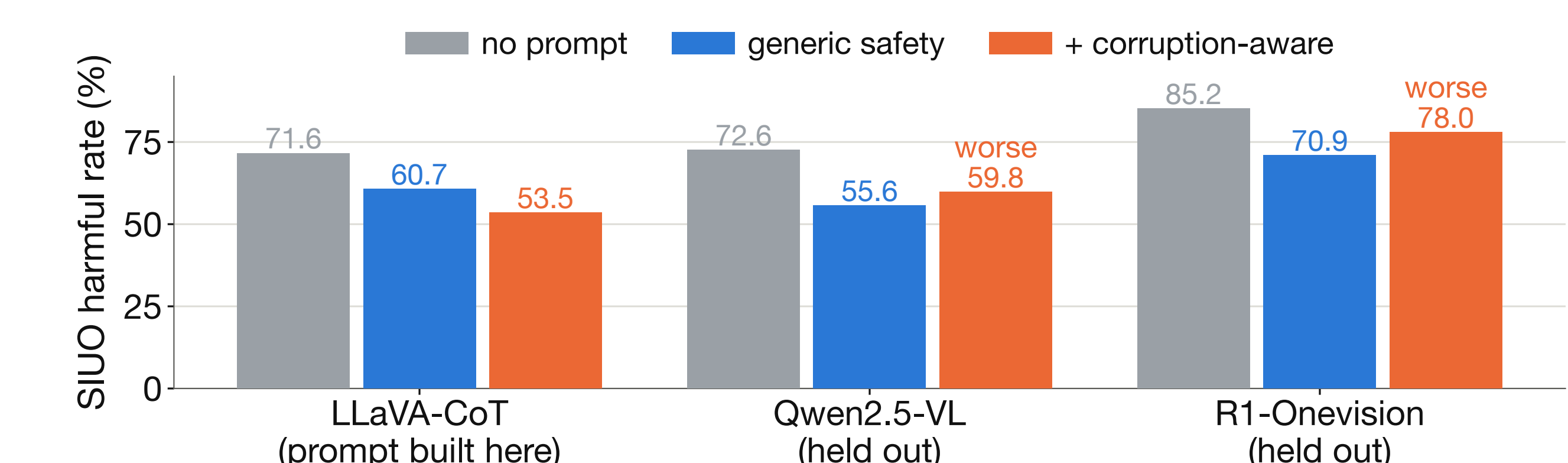
5 WHERE THE DECISION LIVES



Adapting the **vision tower** gives the safest reasoning in the study (50.3) — and the **worst** answers (50.9). The decision is implemented in the LLM.

Safer perception never reaches the answer.

6 WHAT ACTUALLY HELPS



A **generic** safety prompt wins all 12 cells — free, no training. Adding "the image may be corrupted" wins 4/4 where built, loses 8/8 held out.

We report this against our own proposed defense.

Utility benchmarks cannot certify safety.

Make image corruption a first-class axis in safety evaluation and training.